

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib.
1	Soil Sensor Live Telemetry Ingestion	Ingests real-time stream matrix parameters including NPK levels, moisture percentage, and ambient ground temperatures.	Data Ingestion Hub	Done	1.0
2	Automated Weather Forecast Integrator	Polls external meteorological APIs to map hyper-local precipitation projections with crop growth milestones.	API Integration Module	Done	0.5
3	Crop Yield Prediction Engine	Machine learning algorithm assessing historic crop performance baselines against real-time micro-climate feeds.	ML Analytics Core	Done	1.5
4	Dynamic Irrigation Scheduling	Computes daily evapotranspiration metrics to automate optimal water distribution recommendations.	Backend / Decisions Logic	Done	0.5
5	Interactive Yield Analytics Dashboard	A visual workspace offering timeline matrix plots, metric toggles, and multi-format reporting exports.	User Interface (UI)	Done	0.5
6	Automated Disease & Pest Risk Alerting	Evaluates critical humidity spikes against specific threshold indices to trigger real-time mitigation push flags.	Backend / Alerting System	Done	0.5
7	Secured OAuth2 Farmer Authentication	Provides role-based secure login tokens for field operators, regional administrators, and research stakeholders.	Security Module	Done	0.5
8	Offline Edge Synchronization Sync	Enables local data caches to record sensor matrix values during field network blackout conditions.	Database / Sync Storage	In Progress	0.0

Feature Summary:

Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1 Feature	Interactive Yield Analytics Dashboard Workspace
2	Backend/Logic	3 Features	Crop Yield Prediction Engine, Dynamic Irrigation Scheduling, Alerting
3	Database / Storage	2 Features	Offline Edge Data Sync, Live Sensor Telemetry Historical Log Store
4	API / Integration	1 Feature	Automated Weather Forecast API Connector Wrapper
5	Security / Authentication	1 Feature	Secured User Validation Token Protocol (OAuth2 Encryption)